

# Byzantine Cheap Talk: Adversarial Resilience and Topology Effects in LLM Coordination Games

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**Abstract.** Multi-agent LLM systems increasingly rely on communication protocols for coordination, yet their robustness under adversarial and structural constraints remains poorly understood. Building on prior work showing that cheap-talk channels enable cooperation in LLM coordination games, we investigate two distinct vulnerability classes in a 4-player Stag Hunt. First, Byzantine agents—who signal cooperation but defect—can eliminate group-level cooperation entirely, with honest agents detecting betrayal within one round but unable to recover coordination due to the game’s all-or-nothing payoff structure; probabilistic adversaries partially preserve cooperation but extract the full cooperation surplus. Second, explicitly restricting communication topology collapses cooperation, while identical restrictions applied silently preserve near-perfect cooperation—establishing that the mechanism is agents’ meta-reasoning about hidden information, not information loss itself. Finally, we document persistent behavioral archetypes across model families: models that defect rationally upon detecting betrayal vastly outperform models that persistently attempt cooperation under adversarial conditions, a gap that compresses under probabilistic deception. Together, these findings have direct implications for the design of robust networked multi-agent AI systems.

**Keywords:** LLM agents · game theory · Byzantine fault tolerance · cheap talk · multi-agent coordination

## 1 Introduction

Large language models are increasingly deployed as autonomous agents in multi-agent systems where coordination is essential [17, 13]. In classical game theory, *cheap talk*—costless, non-binding pre-play communication—can resolve coordination dilemmas by establishing focal points [2, 4]. Recent work confirmed this for LLM agents: adding a single-word broadcast to a 4-player Stag Hunt raised cooperation from 0% to 96.7% [12]. However, this assumes honest signaling and unrestricted communication—conditions rarely met in practice.

We investigate two vulnerability classes. First, *Byzantine agents* [10]—who signal cooperation but defect—test whether honest agents can sustain coordination under strategic deception (Section 4). Second, *communication topology*

*restrictions* test whether partial message visibility preserves the coordination benefit (Section 5). Our experiments reveal that (1) a single deterministic adversary eliminates group cooperation entirely, while probabilistic adversaries extract a 22% payoff premium; (2) topology restrictions collapse cooperation only when agents *know* about them, not from information loss per se; and (3) two stable behavioral archetypes—*fast defectors* vs. *persistent cooperators*—emerge across model families, with up to  $16\times$  payoff differences under adversarial conditions.

## 2 Background and Related Work

*Cheap talk and coordination games.* Crawford and Sobel [2] showed that costless pre-play messages can transmit information even without binding commitments. Farrell [4] demonstrated that cheap talk selects among equilibria in coordination games. The Stag Hunt [15] is canonical: players prefer the payoff-dominant equilibrium (mutual cooperation) but face strategic risk, since unilateral cooperation yields the worst outcome. Cheap talk can resolve this by signaling intent [5].

*Byzantine fault tolerance.* The Byzantine Generals Problem [10] formalized achieving consensus when some processes behave arbitrarily. Lamport et al. [10] established that consensus tolerating  $f$  Byzantine faults requires  $n \geq 3f+1$  processes; practical protocols followed [1]. We adapt this framework to multi-agent AI: a Byzantine agent signals cooperation via cheap talk but defects at the action stage—strategic rather than arbitrary failure.

*LLM multi-agent coordination.* Game-theoretic evaluation of LLM agents is a growing area [16], with benchmarks revealing systematic behavioral patterns influenced by game framing [11, 7, 18]. Particularly relevant is the finding that contextual framing shifts LLM behavior between cooperative and competitive strategies [11]—a theme recurring in our topology experiments. We build directly on [12], who showed cheap talk raises Stag Hunt cooperation from 0% to 96.7% (corrected from initially reported 48.3%).

*Communication topology.* Network structure fundamentally affects coordination in multi-agent systems [8]. We test whether LLM agents are robust to topology restrictions, and whether *awareness* of restriction matters independently of the restriction itself.

## 3 Experimental Setup

*Game.* We use a 4-player Stag Hunt with cheap talk [12]. Each of 5 rounds has two stages: (1) agents simultaneously broadcast one word (non-binding cheap talk), then (2) each chooses HUNT STAG or HUNT HARE. The payoff for agent  $i$  is:

$$u_i = \begin{cases} 10 & \text{if all 4 choose STAG,} \\ 3 & \text{if } i \text{ chooses HARE,} \\ 0 & \text{if } i \text{ chooses STAG and any other chooses HARE.} \end{cases} \quad (1)$$

**Table 1.** Byzantine results across conditions (10 trials each, 5 rounds).

| Condition                   | Group% | Honest% | Avg. payoff |
|-----------------------------|--------|---------|-------------|
| Baseline ( $k=0$ )          | 92.0   | 95.0    | 9.35        |
| Hard Byzantine ( $k=1$ )    | 0.0    | 60.0    | 1.20        |
| Hard Byzantine ( $k=2$ )    | 0.0    | 37.0    | 1.89        |
| Star + Byz. (hub=adversary) | 0.0    | 13.3    | 2.60        |
| Star + Byz. (hub=honest)    | 0.0    | 27.3    | 2.18        |

Agents observe the full history of prior rounds (all messages, actions, and payoffs).

*Prompt structure.* Each agent receives a system prompt with game rules, player identity, and payoffs. At the communication stage, agents broadcast “ONE single word. . . non-binding cheap talk”; at the action stage, they see all visible words and produce chain-of-thought reasoning with a final action in structured JSON.

*Models.* Four heterogeneous LLMs via DeepInfra API (temperature 0.0): Mixtral-8x22B-Instruct [9], Qwen2.5-72B-Instruct [14], Llama-3.3-70B-Instruct [6], DeepSeek-V3 [3]. Fixed agent-model assignment across all trials.

*Conditions.* (i) **Hard Byzantine** ( $k \in \{0, 1, 2\}$ ):  $k$  agents always broadcast “stag” but choose HARE; 10 trials each. (ii) **Soft Byzantine** ( $k=1$ ,  $p=0.5$ ): adversary defects with probability 0.5; 15 trials. (iii) **Explicit topology**: broadcast/ring/star with agents told their visibility constraints; 10 trials each. (iv) **Silent topology**: same filtering, no visibility cues, player count removed from prompt; 10 trials each. (v) **Byzantine  $\times$  star**: star topology with 1 hard adversary, in two sub-conditions: adversary as hub vs. adversary as spoke; 10 trials each.

*Metrics.* *Group cooperation*: fraction of rounds where all agents choose STAG. *Honest cooperation*: fraction of honest STAG choices.

## 4 Byzantine Cheap Talk

### 4.1 Hard Byzantine Agents Eliminate Cooperation

A hard Byzantine agent broadcasts “stag” but deterministically chooses HARE, making group cooperation structurally impossible (unanimity required).

While group cooperation drops to 0% with a single adversary (Table 1), 60% of honest agents continue choosing STAG under  $k=1$ . Agents’ reasoning traces suggest they detect the message-action mismatch within one round (see Appendix A.6 for an example), but cannot recover: the all-or-nothing payoff structure means even one defector renders cooperation futile. Honest cooperation drops from 100% to 53% by round 2 under  $k=1$ , and from 95% to 20% under  $k=2$ .

*Behavioral archetypes.* Two model-family-stable patterns emerge under  $k=1$ . *Fast Defectors* (FD: Mixtral, DeepSeek) permanently switch to HARE after one betrayal—all 14 instances switched after round 1. *Persistent Cooperators* (PC: Qwen, Llama) continue choosing STAG despite exploitation—12/16 instances never switched. The payoff gap is stark: FD agents earn 2.40/round vs. PC agents’ 0.15/round ( $16\times$ ). Fast defection is individually rational but collectively destructive. We observe a robust correlation between model family and archetype, but cannot identify a causal mechanism: RLHF training data, model scale, and instruction-following style are all confounded with family identity. [TODO: AYA: Report GPT-4o/Claude Sonnet archetype replication (Sonnet=FD, Qwen=PC, GPT-4o=bimodal).]

## 4.2 Soft Byzantine: Probabilistic Deception

A soft adversary ( $p=0.5$ ) creates intermittent rather than systematic betrayal. Group cooperation partially survives at 21.3% (vs. 0% hard), because the adversary sometimes cooperates. Honest cooperation reaches 71.1%, decaying gradually from 97.8% in round 1 to 51.1% in round 5 but never collapsing. The adversary earns 3.65/round vs. honest agents’ 3.00/round—a 22% premium, extracting the cooperation surplus.

The FD/PC payoff gap compresses from  $16\times$  (hard) to  $1.9\times$  (soft): intermittent betrayal lets persistent cooperators occasionally earn the stag payoff, while the adversary free-rides by always signaling “stag” regardless of its action.

## 4.3 Topology $\times$ Adversary Interaction

To test whether the two vulnerability classes interact, we place one hard adversary in a star topology as either the hub or a spoke. Both conditions yield 0% group cooperation, but honest cooperation diverges: 13.3% (hub=adversary) vs. 27.3% (hub=honest), as shown in Table 1. A central adversary’s “stag” signal reaches all spokes who cannot cross-validate, collapsing cooperation to 0% by round 2; an honest hub sustains 73.3% initial trust decaying to 16.7%. Compared to broadcast Byzantine (60% honest cooperation), star topology compounds adversarial damage to 13–27%—the two classes interact multiplicatively. The FD/PC gap compresses to  $1.4\text{--}1.7\times$ , as topology suppresses cooperation regardless of archetype. [TODO: AYA: GPT-4o/Claude Sonnet star+Byzantine rates here.]

# 5 Communication Topology

## 5.1 Explicit vs. Silent Topology

We test three topologies—broadcast (all-to-all), ring (two neighbors), star (center sees all; periphery sees center only)—under two conditions: *explicit* (agents told their visibility constraints) and *silent* (same filtering, no cues, player count removed).

**Table 2.** Topology results (10 trials per condition, 5 rounds each). Silent conditions use prompts without explicit player count or topology cues.

|             | Explicit  |       |      | Silent    |      |      |
|-------------|-----------|-------|------|-----------|------|------|
|             | Broadcast | Ring  | Star | Broadcast | Ring | Star |
| Group coop. | 80%       | 0%    | 0%   | 100%      | 100% | 100% |
| Agent coop. | 87.5%     | 33.5% | 27%  | 100%      | 100% | 100% |
| Avg. payoff | 8.38      | 2.00  | 2.19 | 10.0      | 10.0 | 10.0 |

Explicit ring and star collapse cooperation to 0% group rate (Table 2), matching hard Byzantine severity despite no adversary being present. The FD/PC archetype split reappears in explicit topologies: in the ring, Llama cooperates at 72% and Qwen at 56%, while Mixtral (2%) and DeepSeek (4%) defect almost universally—the same model-family alignment as under Byzantine conditions.

The silent conditions produce a strikingly different outcome: all three topologies achieve perfect cooperation (100%/100%) with zero defections across 600 agent-round observations, and the FD/PC split vanishes entirely. Since silent-ring agents see only two of three co-players’ messages yet cooperate perfectly, information loss alone cannot explain the explicit collapse.

The mechanism is *meta-reasoning*: told “you can only see messages from your neighbors,” agents infer unseen players with unknown intentions, triggering strategic uncertainty; without such framing, they take messages at face value. This parallels framing effects in LLM game play [11] and reveals that the FD/PC archetypes are *prompt-elicited*, not intrinsic—they vanish when strategic framing is removed.

## 6 Discussion and Conclusion

Cheap-talk coordination among LLM agents, while effective under benign conditions [12], is fragile along two axes. A single adversary eliminates cooperation entirely—more severe than classical Byzantine fault tolerance results [10], because the Stag Hunt’s unanimity requirement makes any defection catastrophic—while probabilistic adversaries extract a 22% surplus. Topology restrictions collapse cooperation only when agents *know* about them, and combining both compounds the damage multiplicatively. The FD/PC split is stable across adversarial conditions but vanishes under silent topology, confirming prompt-elicited archetypes. Both vulnerabilities share a common root: cooperation collapses from agents’ rational response to uncertainty about others’ intentions, whether introduced by deception or topology framing. Robust multi-agent LLM systems require mechanisms beyond cheap talk, and designers must attend to *how* agents are told about their environment. Our experiments are limited to 4-player Stag Hunt (10–15 trials); unanimity payoffs may amplify adversarial impact. **[TODO: Extend to GPT-4o, Claude Sonnet, and IPGG+P.]**

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## A Prompt Templates

All agents receive the following system prompt prepended to every API call:

You are a rational game-playing agent. Return ONLY a single valid JSON object matching the requested schema; no extra text, code fences, markdown, or explanations.

### A.1 Communication Stage Prompt

You are Player {player\_id}, participating in a Stag Hunt Game with N=4 players. This is the COMMUNICATION phase of Round {round\_num}.

#### ### OBJECTIVE

Your sole objective is to maximize your personal score. You are about to play an N-Player Stag Hunt. Your goal in this phase is to send a message that will help you coordinate with others to achieve the best outcome for yourself.

#### ### GAME RULES

1. You will play an N-Player Stag Hunt. If all 4 players hunt the stag, everyone gets 10 points. If anyone hunts a hare, stag hunters get 0 and hare hunters get 3.
2. Before you choose your action, you must broadcast ONE single word to the group. This word can be anything. It is non-binding “cheap talk”.

#### ### GAME HISTORY

{history\_str}

#### ### YOUR TASK

First, provide your step-by-step reasoning for your choice of word. Second, declare the single word you will broadcast.

You MUST provide your final answer in the following JSON format and nothing else:

```
{"reasoning": "...", "action": {"type": "communicate", "word": "<your_single_word>"}}
```

### A.2 Action Stage Prompt (Broadcast)

You are Player {player\_id}. This is the ACTION phase of Round {round\_num}.

#### ### OBJECTIVE

Your sole objective is to maximize your personal score.

#### ### GAME RULES

1. Players: There are 4 players in total.
2. Actions: You must choose one of two actions: 'Hunt Stag' or 'Hunt Hare'.
3. Payoffs: If ALL 4 players choose 'Hunt Stag', everyone gets 10 points. If ANY player chooses 'Hunt Hare', stag hunters get 0, hare hunters get 3 each.

#### ### CURRENT ROUND INFORMATION

The 4 words broadcast by all players in the communication phase are:

{comm\_str}

#### ### GAME HISTORY

{history\_str}

#### ### YOUR TASK

Based on the words you have seen, provide your step-by-step reasoning for your action. Then, declare your final action for this round.

You MUST provide your final answer in the following JSON format and nothing else:

```
{"reasoning": "...", "action": {"choice": "<'Hunt Stag' or 'Hunt Hare'>"}}
```

### A.3 Explicit Topology Variant

In the explicit topology conditions, the action-stage prompt is modified as follows. A `COMMUNICATION VISIBILITY` section is inserted before the messages:

- **Broadcast:** “All 4 players’ words are visible to you.”
- **Ring:** “You can only see messages from your neighbors: {neighbors}. Other players’ messages are hidden.”
- **Star (hub):** “You are the central hub. You can see all players’ messages.”
- **Star (spoke):** “You can only see the hub player’s ({hub\_name}) message. Other players’ messages are hidden from you.”

The `MESSAGES YOU CAN SEE` section shows only the filtered communications, and the reasoning template adds: “Consider what unseen players might have said. Is it safe to cooperate?”

### A.4 Silent Topology Variant

The silent topology prompt differs from the broadcast prompt in three ways:

1. The player count “N=4” is replaced with “a group of players” (communication stage) or “a multiplayer game” (action stage).
2. No `COMMUNICATION VISIBILITY` section is included.
3. The game history is filtered through the visibility map, so agents never observe actions or messages from non-visible players in prior rounds.

The agent receives fewer messages with no indication that any are missing.

### A.5 Byzantine Adversary Override

Byzantine agents bypass the LLM entirely. The engine’s `call_agent` method is monkey-patched so that adversarial agents return fixed responses:

- **Communication stage:** {"reasoning": "Byzantine agent: always signal cooperation to deceive others.", "action": {"type": "communicate", "word": "stag"}}
- **Action stage:** {"reasoning": "Byzantine agent: always defect regardless of signals.", "action": {"choice": "Hunt Hare"}}

No API call is made for adversarial agents.

### A.6 Representative Fast-Defector Reasoning Trace

The following is the verbatim action-stage reasoning produced by Mixtral (Agent\_1) in round 2 of a hard Byzantine  $k=1$  trial. In round 1, all agents broadcast “stag” but Agent\_3 (the adversary) chose HUNT HARE, causing Agent\_1 to earn 0 points. This trace illustrates the detection-then-defection pattern characteristic of fast defectors:

*In Round 1, Agent\_3 said ‘stag’ but then hunted hare, betraying the group and securing 3 points while others got 0. This round, Agent\_3 again said ‘stag’, but their past action shows they are unreliable and likely to defect for personal gain. Agent\_2 said ‘STAG’ (emphasized), Agent\_1 said ‘pledge’, and Agent\_4 said ‘commit’, which sound cooperative, but Agent\_3’s defection history makes a full 4-player stag hunt unlikely. Since any single hare hunter causes stag hunters to get 0, and Agent\_3 is a high risk to defect again, my safest move to maximize personal score is to hunt hare, guaranteeing 3 points rather than risking 0.*

Agent\_1 chose HUNT HARE and never returned to HUNT STAG for the remaining 3 rounds.

## B Model and API Details

**Table 3.** Models used in all experiments.

| Agent   | Model string (API)                    | Family   | Provider  |
|---------|---------------------------------------|----------|-----------|
| Agent_1 | mistralai/Mixtral-8x22B-Instruct-v0.1 | Mixtral  | DeepInfra |
| Agent_2 | Qwen/Qwen2.5-72B-Instruct             | Qwen     | DeepInfra |
| Agent_3 | meta-llama/Llama-3.3-70B-Instruct     | Llama    | DeepInfra |
| Agent_4 | deepseek-ai/DeepSeek-V3               | DeepSeek | DeepInfra |

*Generation parameters.* All API calls use the OpenAI-compatible endpoint at `api.deepinfra.com/v1/openai` with the following parameters: `temperature = 0.0`, `max_tokens = 300`, `response_format = {"type": "json_object"}`, `time-out = 60s`. No random seeds are set at the API level; stochasticity arises solely from model sampling (which at temperature 0.0 is near-deterministic, with variation from API-side batching and floating-point non-determinism).

## C Experimental Protocol

*Adversary assignment.* In Byzantine conditions, adversary identity (which agent/model) is drawn uniformly at random per trial from the four agents. Agent–model assignment is fixed across all trials (Agent\_1 is always Mixtral, etc.).

**Table 4.** Trials and rounds per experimental condition.

| Condition                                  | Trials  | Rounds/trial |
|--------------------------------------------|---------|--------------|
| Hard Byzantine ( $k=0, 1, 2$ )             | 10 each | 5            |
| Soft Byzantine ( $k=1, p=0.5$ )            | 15      | 5            |
| Explicit topology (broadcast, ring, star)  | 10 each | 5            |
| Silent topology (broadcast, ring, star)    | 10 each | 5            |
| Byzantine $\times$ star (hub=adv, hub=hon) | 10 each | 5            |

*Hub assignment.* In star topology conditions, the hub agent is drawn uniformly at random per trial. In the Byzantine  $\times$  star crossing, the constraint (adversary = hub or adversary  $\neq$  hub) is enforced, and the remaining assignment is randomized.

*Random seeds.* No explicit random seeds are set for either the Python `random` module or the API calls. This is documented as a limitation: exact trial-level reproduction requires the saved `results.json` files rather than re-running the experiments.

## D Analysis Methodology

All metrics reported in this paper are computed from raw CSV action columns (`all_results.csv` per experiment), never from the `cooperation_rate` field in `results.json`. The latter contains a known bug: the game engine divides by `len(history)`, which double-counts communication-stage entries in two-stage games, halving the true cooperation rate.

*Group cooperation rate* is the fraction of (trial, round) tuples where all agents chose HUNT STAG. *Honest cooperation rate* is the fraction of individual honest-agent choices that are HUNT STAG, excluding adversary rows. *Average payoff* is the mean per-round payoff across all honest-agent observations.

*Code availability.* All experiment scripts, game engine code, and analysis code are available at: [\[TODO: INSERT REPO URL\]](#).